

DatOps

Mandi-to-Market Supply Chain Optimizer

Turn messy mandi, price, logistics and weather data into an operator-ready command center.

DATA RESCUE

clean + standardize

ANALYTICS

metrics + signals

DECISIONING

risk + action

AGRIQUERY

ask in plain English

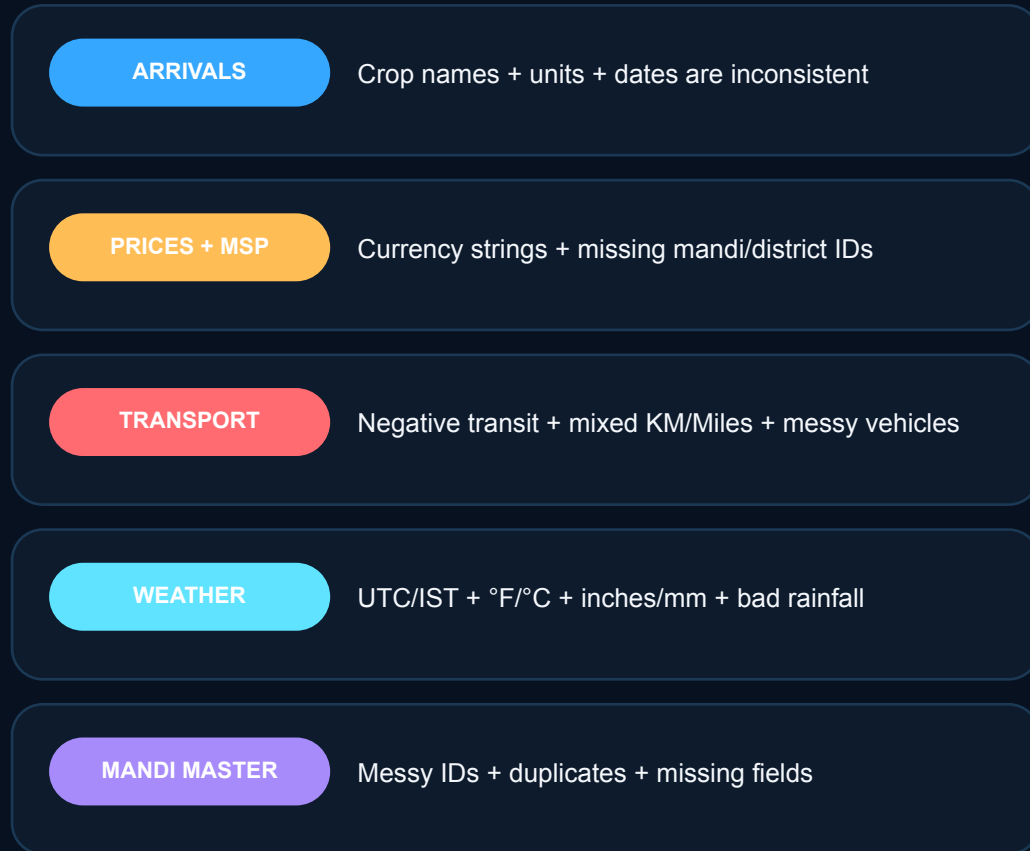
CORE STORY

“What is happening?” → “Where is the pressure?” → “What should an operator investigate?”

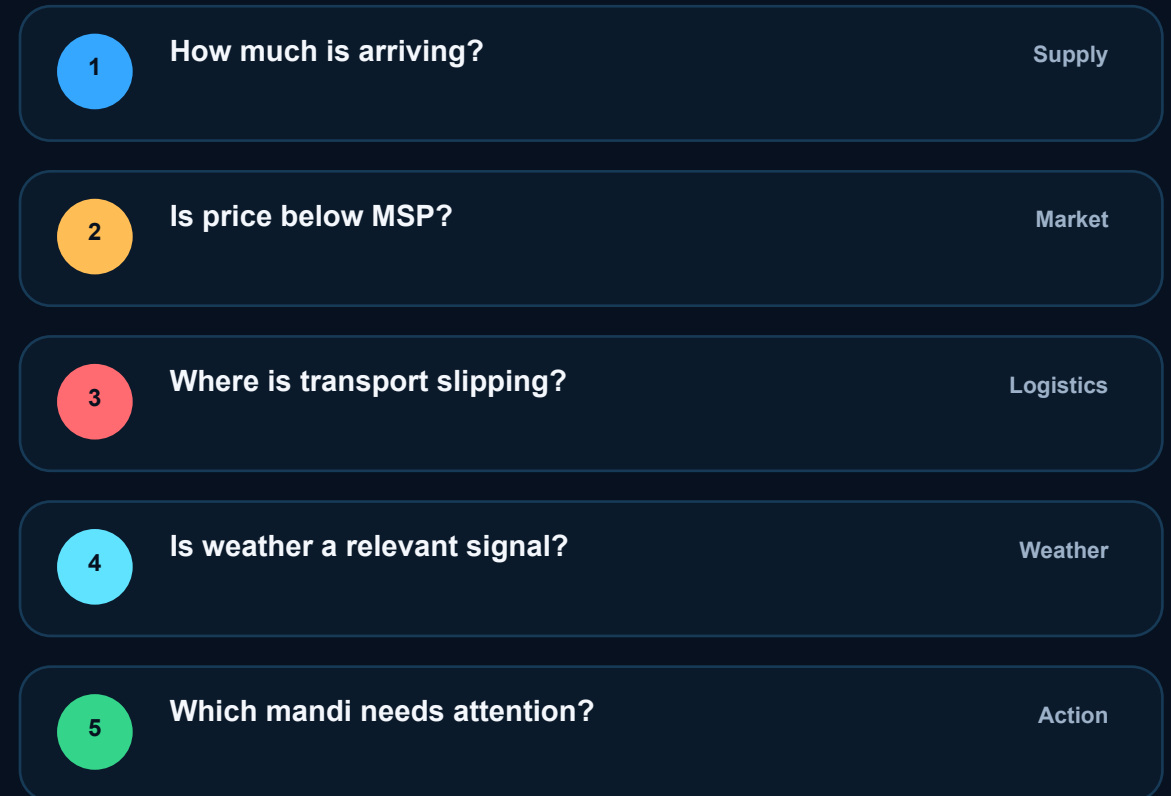
Public dashboard • public GitHub • reproducible DuckDB pipeline

The data is messy. The decision still has to be fast.

The State Agriculture Board needs one place to read supply, market, logistics and weather signals without hand-cleaning five source files.



THE DECISION LOOP



DESIGN PRINCIPLE clean first • calculate once • reuse everywhere

From five messy files to one trusted analytical layer

Keep the source visible, make analytical usability explicit.

SOURCE DATASETS

5

arrivals, price, transport, weather, master

RAW RECORDS

63,210

before any analytical filtering

CROP VARIANTS

36 → 6

canonical crops

ARRIVAL DUPLICATES

750

exact duplicate rows

TRANSPORT DUPLICATES

400

exact duplicate rows

WHAT NEEDED RESCUING

Missing arrival units 5,143

Negative transit hours 563

Negative rainfall values 1,518

Missing price mandi IDs 1,235

Status fields preserve auditability:

quantity_status

price_status

transit_status

THE CHOICE

Do not silently throw away every unusual value.

Instead: normalize what can be repaired, flag what is impossible, and let each metric decide what is analytically valid.

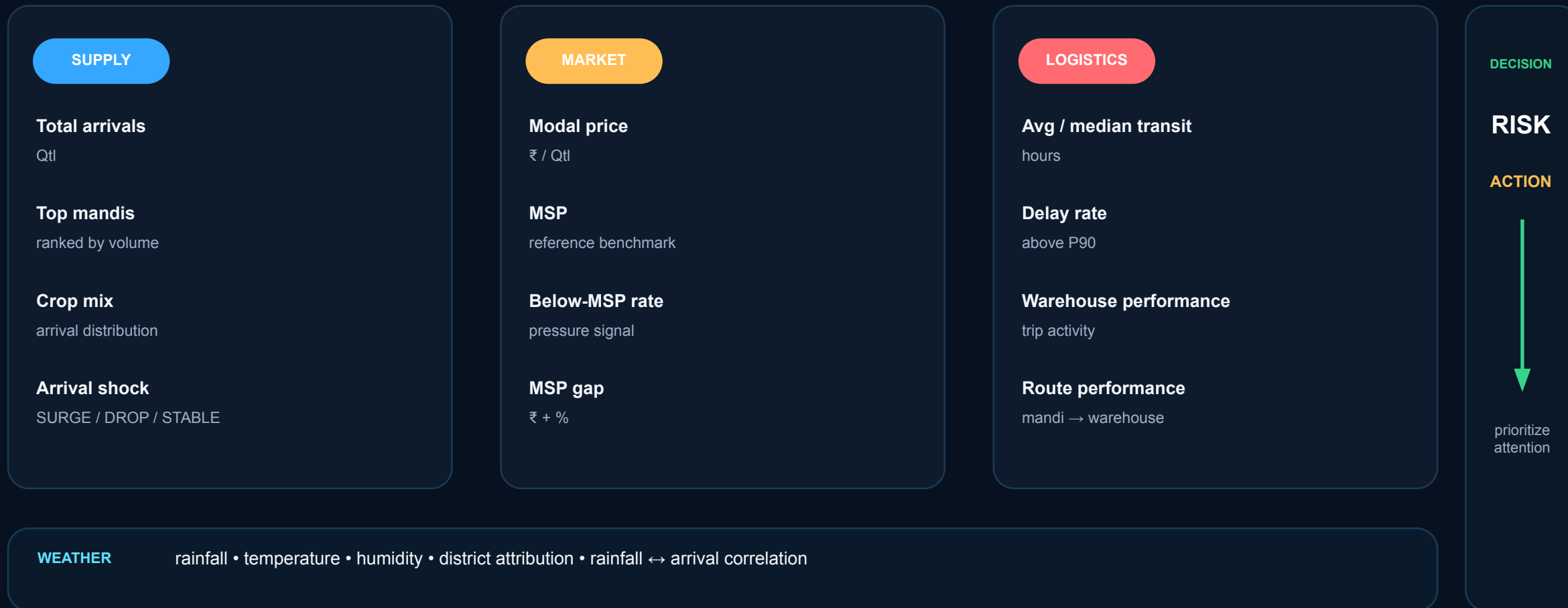
Examples

1 Tonne → 10 Qtl; 1 Qtl → 100 KG

Miles → KM; °F → °C; inches → mm

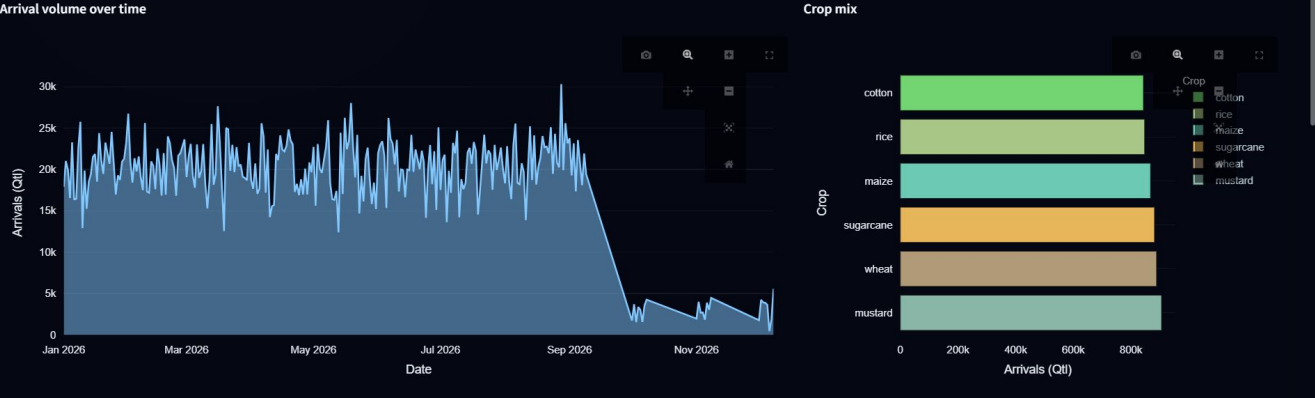
Business metrics are domain-specific — and fully implemented

The dashboard uses agricultural supply-chain KPIs, not generic SaaS metrics.

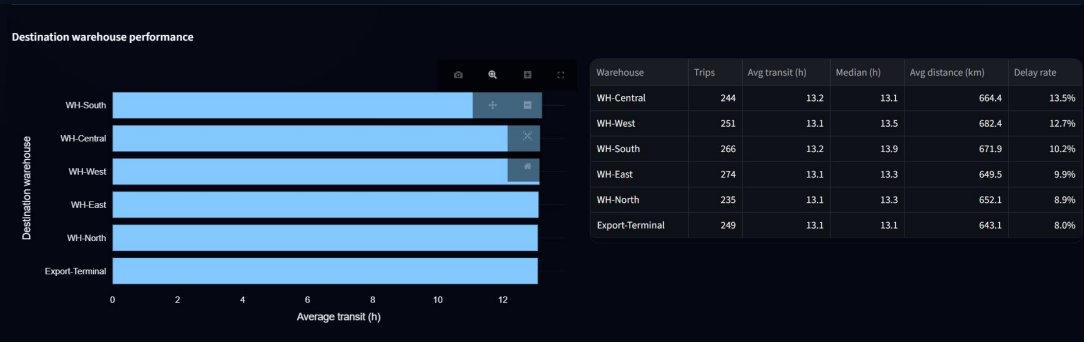
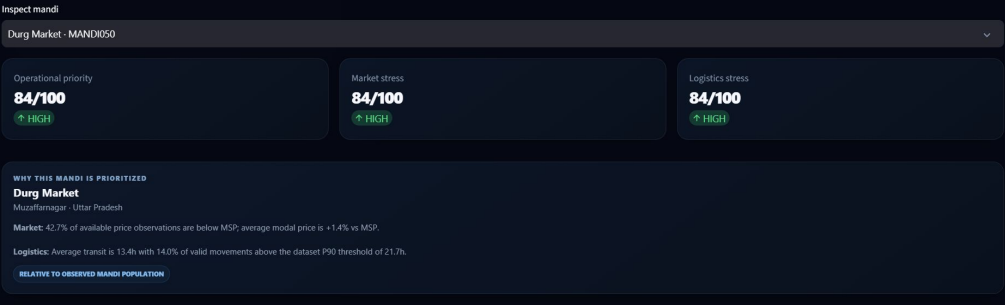


Dashboard insights

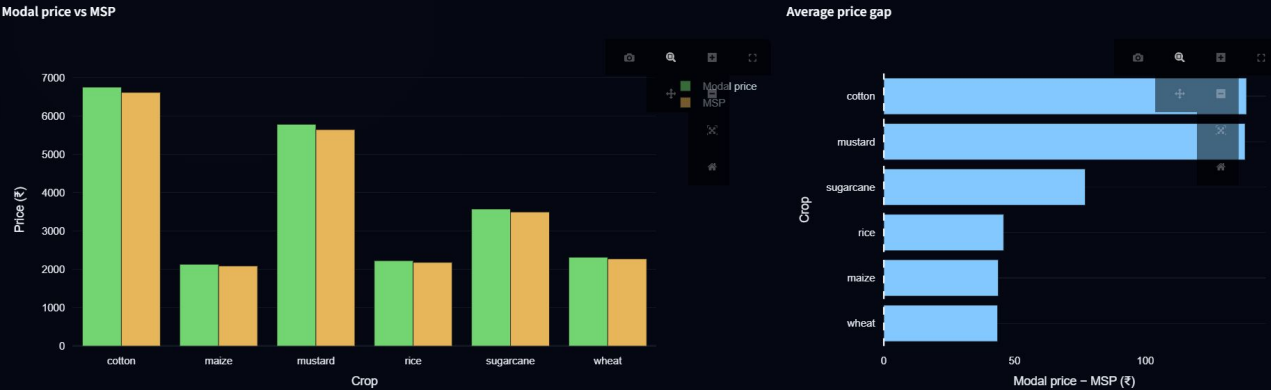
Supply intelligence



Explainable risk intelligence



Price intelligence



Mandi-to-Market Command Center

One command center for supply, market pressure, logistics, weather and risk.



DatOps
MANDI INTELLIGENCE

Operational analytics for supply, markets, logistics and risk.

CONTROL FILTERS

State

All states

District

All districts

Mandi

All mandis

Crop

All crops

ANALYSIS WINDOW

OverviewSupplyPriceLogisticsWeatherMandi ExplorerRiskAgriQuery

Deploy

DATOPS · MANDI INTELLIGENCE



Mandi-to-Market Command Center

A decision cockpit for agricultural supply, price pressure, logistics performance and weather signals — with every view backed by a validated analytical data layer.

Analytics Engine

DuckDB

Connected

Total arrivals

5,234,458 Qtl

Avg modal price

₹3,794

Below MSP

40.1%

Avg transit

13.1 h

Delay rate

10.5%

Executive signals

A fast operational read before drilling into the tabs.

Price pressure

40.1% below MSP

3,824 of 9,541 valid price observations are below the available MSP.

Logistics pressure

10.5% delayed

Transport records above the empirical P90 threshold of 21.7 hours.

Analytical coverage

20,394 arrival rows

11,921 price rows and 1,519 valid transport rows are in scope.

DATOPS | MANDI INTELLIGENCE

LIVE STREAMLIT

Public app
deployed
6

From KPIs to operational priority

Arrival Shock + market pressure + logistics pressure + weather signals turn monitoring into triage.

RISK ENGINE

MARKET

70% below-MSP pressure + 30% gap

LOGISTICS

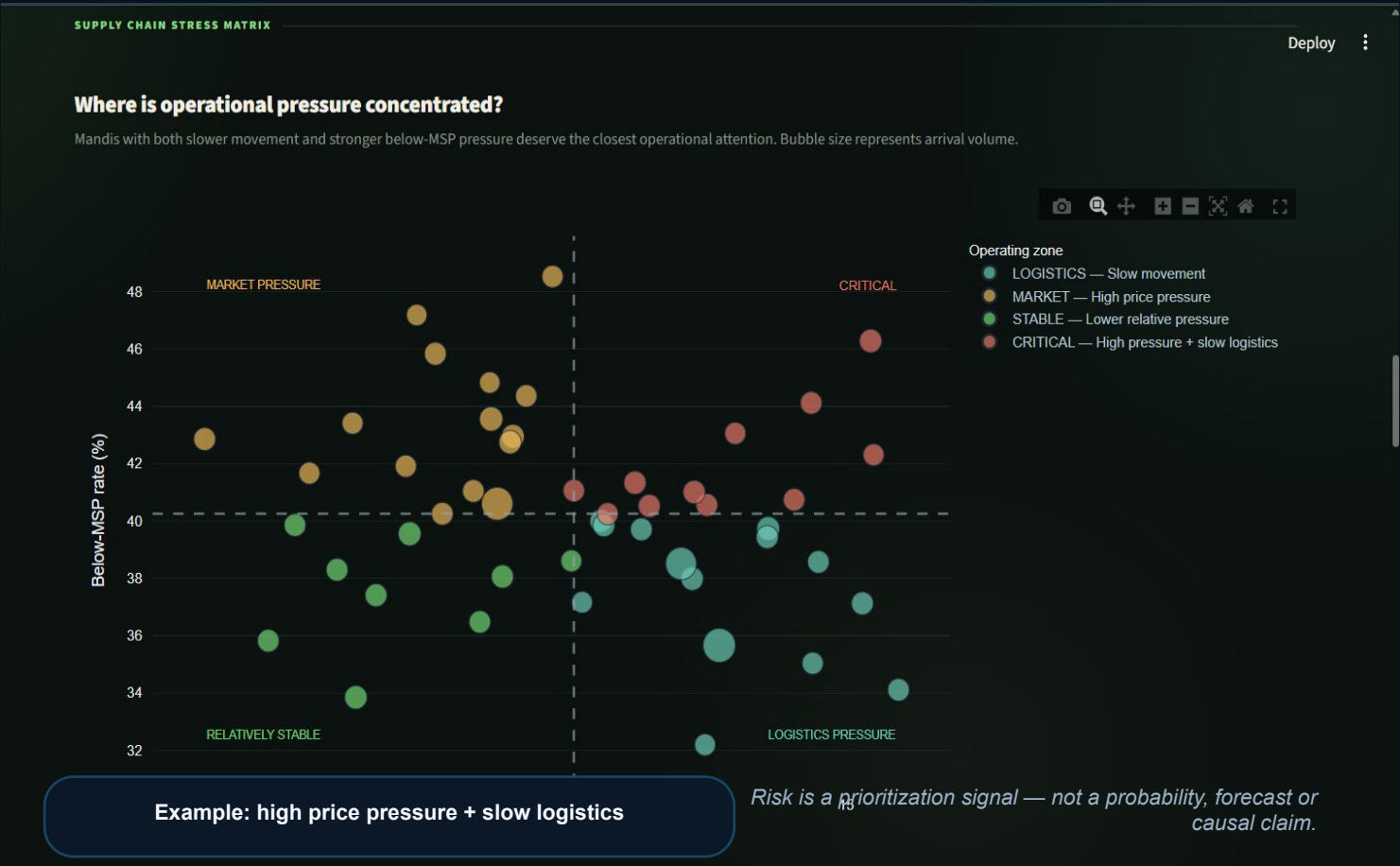
60% transit + 40% delay rate

WEATHER

70% rainfall intensity + 30% rainy-sensor share

MANDI RISK

market + logistics → operational priority



AgriQuery: ask normally, get the right analytical view

Natural language → intent → validated query plan → chart → written explanation.


DatOps
MANDI INTELLIGENCE

CONTROL PANEL

Location scope

State

All states

District

All districts

Mandi

All mandis

Crop

All crops

Analysis window

Arrival dates

2026/01/01 – 2026/12/08

Overview

Supply

Market

Logistics

Weather

Mandis

Risk

AgriQuery

Share

☆

✎

🔄

⋮

Ask AgriQuery

Ask in plain language. AgriQuery resolves entities, selects the analytical path, retrieves grounded evidence, chooses a visual and explains the result.

QUESTION – INTENT – DATA – VISUAL – DECISION

Load this question into AgriQuery

Top 5 mandis by arrivals

Which mandis are below MSP?

What is the risk for MANDI047?

Which mandis have both high ...

Show wheat arrivals for the las...

ASK

Natural-language question

GROUNDING

Validated DuckDB

OUTPUT

Visual + explanation

Show wheat arrivals for the last 30 days

▶ Run analysis

Clear

Validated DuckDB · visual + explanation · no fabricated observations

AgriQuery result

Detected intent

Daily Arrivals

Recommended view

Line

Manage app

THREE BONUS BOXES

01

UNDERSTAND

Natural-language intent

02

VISUALIZE

Right chart for the question

03

EXPLAIN

Summary alongside result

EXAMPLES

- “Top 5 mandis by arrivals” → bar chart
- “What is the risk for MANDI047?” → indicator
- “Daily wheat trend...” → line + MSP reference

AgriQuery: ask normally, get the right analytical view

>>OverviewSupplyMarketLogisticsWeatherMandisRiskAgriQueryShare☆🔗⋮

ASK

Natural-language question

GROUNDING

Validated DuckDB

OUTPUT

Visual + explanation

Which warehouse has the highest average transit time?

▶ Run analysis

Clear

Validated DuckDB · visual + explanation · no fabricated observations

AgriQuery result

Detected intent

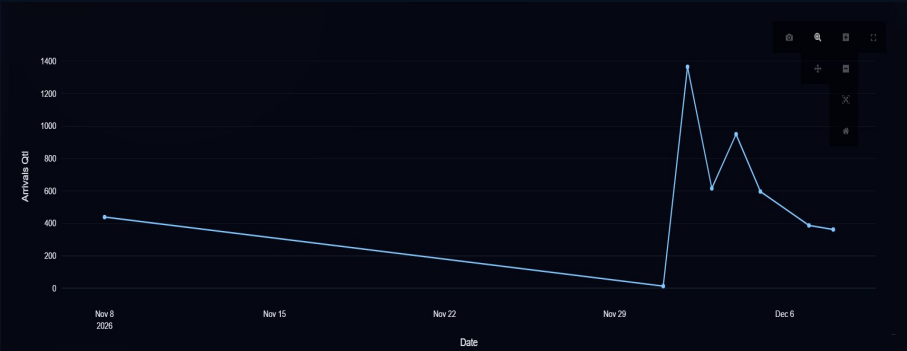
Warehouse Logistics

Recommended view

Bar

● Agent explanation

Returned 6 observations. Shows inbound trip activity by destination warehouse. 'Trips' is used instead of tonnage because the supplied transport records do not contain shipment quantity. Delay uses the empirical P90 transit threshold.



destination_warehouse	trips	avg_transit_hours	median_transit_hours	avg_distance_km	delayed_trips	delay_rate_pct
WH-Central	1557	12.95	12.9	650	162	10.4
WH-North	1611	13	12.9	648.51	163	10.12
Export-Terminal	1519	12.89	12.9	643.32	149	9.81
WH-South	1585	12.86	12.9	647.58	155	9.78
WH-West	1552	13.01	13.15	658.84	146	9.41
WH-East	1495	13.03	13.1	659.8	139	9.3

The official query — without fabricating missing observations

Wheat + Amritsar + last 30 days demonstrates how the agent handles sparse, imperfect data.

“Plot the daily arrival trend of Wheat in Amritsar mandi vs MSP for the last 30 days.”

OBSERVED DAYS

2

only days with valid data

LATEST ARRIVAL

90.62 Qtl

Dec 05

MSP REFERENCE

₹2,275

available price observations

INTERPRETATION

Amritsar = district

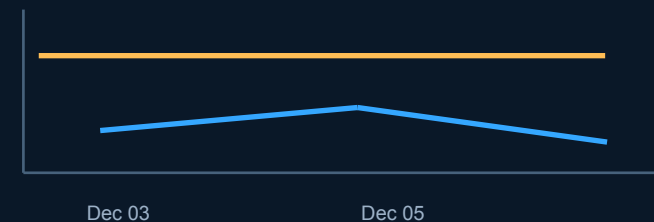
no exact mandi named Amritsar in master

WHY THIS MATTERS

The agent does not invent a continuous daily price series.

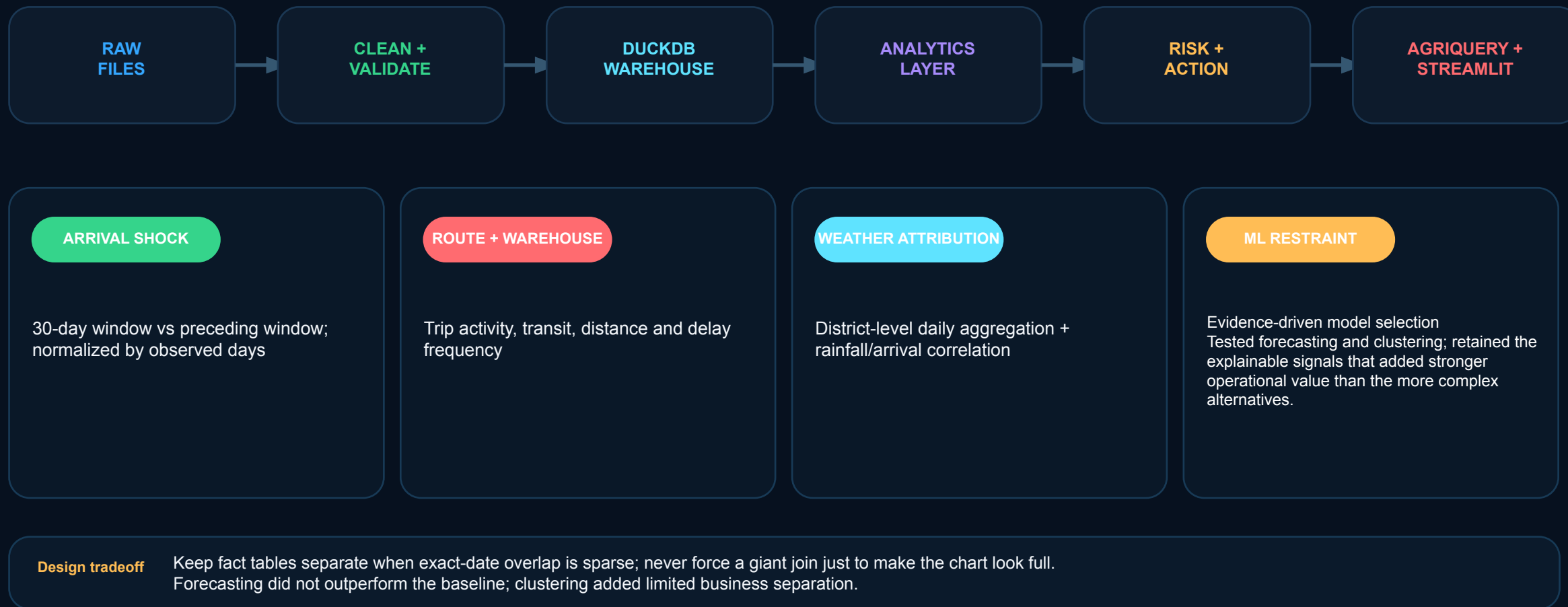
- Interprets “Amritsar” using the available master geography.
- Shows MSP as a reference line from available observations.
- Leaves missing daily values missing instead of fabricating them.

ACTUAL RESULT



A reproducible pipeline — plus the decisions we intentionally did not fake

Modular code, validated data, safe query routing, and explainable signals.



From messy source data to an operator-ready decision loop.

DatOps is built to answer the questions that matter on the ground:

SUPPLY

What is arriving?

MARKET

Where is price pressure?

LOGISTICS

Where are routes slipping?

RISK

Which mandi needs attention?

ACTION

What should be investigated?

LIVE DEMO

datops-mandi2market.streamlit.app

GitHub

github.com/srkk05/DatOps

TRACK 3 • AGRITECH

Clean data. Defensible metrics. Actionable intelligence.

THANK YOU